

The block-rank theorem for Fermat CRT blocks

The per-block rank of the Chinese-Remainder splitting of a Fermat cell is a Degtyarev-Shimada rank polynomial — equivalently, the constant term of a closed lattice walk — for every block dimension, proved by a tensor identity from the Degtyarev-Shimada source

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The capstone of the campaign's structural results. [THE_LOCALIZATION_THEOREM.md](#) proves *where* any torsion of a Fermat cell must live (one CRT block). [THE_BLOCK_DECOMPOSITION.md](#) measures *how* the block ranks multiply (a product of recursion ladders). [THE_WITHIN_PAIR_FUNCTOR.md](#) opens the *interior* of one block. [THE_CLOSED_WALK_LAW.md](#) names the *value* each ladder rung takes (a closed walk). **This note closes the ladder: it proves that the rank a block contributes at every coupling depth equals that Degtyarev-Shimada value, for every block dimension, by a module-dimension identity read from the Degtyarev-Shimada paper itself.** With it, the block-rank structure of the splitting is a theorem rather than a measured pattern.

0.1 Abstract

Let (n, m) be an even-dimensional Fermat cell and $p \mid m$ a prime. Over F_p the cyclotomic quotient splits, by the Chinese Remainder Theorem, into blocks; the rank a block contributes is governed by the recursion

$$\text{rec}(d, 0) = 1, \quad \text{rec}(d, 1) = d, \quad \text{rec}(d, l) = \text{DS}_{\{2(l-1)\}}(d+1) \quad \text{for } l \geq 2,$$

measured byte-exact across the campaign (the General Product Law, [THE_BLOCK_DECOMPOSITION.md](#)), where $\text{DS}_n(m)$ is the Degtyarev-Shimada complex rank polynomial [2, Remark 4.4]. This note proves the identity $\text{rec}(d, l) = \text{DS}_{\{2(l-1)\}}(d+1)$ holds **for every d** , not only the cases measured. The proof is a chain of module-dimension identities read from the Degtyarev-Shimada source, in three links:

1. **The value is a closed walk.** $\text{DS}_{\{2s\}}(m)$ is the constant term of a closed lattice walk of $2s+2$ steps on $\mathbb{Z}^{\lfloor (m-1)/2 \rfloor}$ — directional steps with rest permitted iff m is even, the rest permission being exactly the published parity correction δ_m . Proved and gate-verified byte-exact in [THE_CLOSED_WALK_LAW.md](#) (Eslabón 4).
2. **The measured block is a Degtyarev-Shimada core module.** By the Degtyarev-Shimada tensor decomposition $\tilde{C}_{\{J_s\}}(2d) = \tilde{C}_{\{J(2s)\}}(2s) \otimes_{\mathbb{Z}} \tilde{S}(s, d)$ [2, Cor 1.7], with the tail $\tilde{S}(s, d)$ a free $\mathbb{Z}$ -module, the rank of the block at depth $l = s+1$ equals $\dim_{\mathbb{C}}$ of the core $\tilde{C}_{\{J(2s)\}}(2s)$ exactly — a free-module dimension, isolated by the clean tensor split. And $\dim_{\mathbb{C}}$ of that core is the closed walk of link 1, by Remark 4.4 applied one dimension down (Eslabón 5).

3. **The step is uniform.** The depth step $l-1 \rightarrow l$ adds two lattice coordinates and is a finite-order holonomic operator for every block dimension (Lipshitz 1988 [4]); this is the structural reason the induction does not blow up as the lattice dimension grows, and it carries the *uniformity* in d , not the equality.

The equality at each depth is therefore standalone — $\text{rec}(d, l) = \dim_{\mathbb{C}}(\text{core}) = \text{walk} = \text{DS}_{2(l-1)}(d+1)$ — and propagates from the base $\text{rec}(d, 1) = d$. **The result is conditional on one identification, named explicitly: that the campaign’s measured CRT all-A block at depth l is the Degtyarev-Shimada core $\tilde{c}_{J(2s)}(2s)$** (the dictionary $e_A \leftrightarrow$ coordinate-zeroing, established earlier in the campaign and confirmed verbatim against [2, Def 1.3, Thm 1.4, §4.6]). The integral torsion question (Degtyarev-Shimada Conjecture 1.2, the campaign’s “swan”) sits outside this chain and remains open: this theorem fixes the per-block **value**, not whether torsion exists. Every numerical statement is reproduced byte-exact by self-contained verifiers; the proof is verified link by link against the source.

0.2 1. Setting and statement

Fix an even-dimensional Fermat cell (n, m) , $n = 2d$, and a prime $p \mid m$. The integral Hodge question for the standard d -spaces reduces, by Degtyarev-Shimada [2], to whether $\dim_{\mathbb{C}}$ of the cycle lattice equals $\dim_{\mathbb{F}_p}$ for every $p \mid m$; the complex rank is the closed-form polynomial $\text{DS}_n(m)$ of [2, Remark 4.4]. Over $\mathbb{F}_p$ the per-variable cyclotomic ring splits, by the Chinese Remainder Theorem, into a $(t-1)$ -primary factor A and a coprime cofactor B ; tensoring over the $n+2$ variables decomposes the cohomology module into CRT blocks. The rank a block contributes at **coupling depth** l (the number of A -pairs in the block) was measured across the campaign to obey

$$\text{rec}(d, l) = \text{DS}_{2(l-1)}(d+1) \quad (l \geq 2), \quad \text{rec}(d, 1) = d, \quad \text{rec}(d, 0) = 1,$$

the **General Product Law** of [THE_BLOCK_DECOMPOSITION.md](#). The mechanism by which the ladders *mesh* as l grows — the coupling — was a measured pattern. This note proves it.

Theorem (block-rank law). *For every $d \geq 0$ and $l \geq 1$, conditional on the dictionary of §2,*

$$\text{rec}(d, l) = \text{DS}_{2(l-1)}(d+1).$$

Equivalently, the rank the block contributes at depth l is the constant term of a closed lattice walk of $2l$ steps on $\mathbb{Z}^{\lfloor d/2 \rfloor}$, with rest permitted iff d is odd.

The published Degtyarev-Shimada closed forms, with $\delta_m = (m-1) \bmod 2$, are $\text{DS}_2(m) = 3m^2 - 9m + 6 + \delta_m$, $\text{DS}_4(m) = 15m^3 - 90m^2 + 175m - 100 + (15m - 39)\delta_m$, and $\text{DS}_6(m) = 105m^4 - 1050m^3 + 3955m^2 - 6335m + 3325 + (210m^2 - 1302m + 2010)\delta_m$; these are the only rows for which [2] prints a polynomial. The theorem covers all $n = 2(l-1)$, i.e. all depths, by the closed-walk reading of link 1 — which is the *definition* of DS_n for $n > 6$, there being no published closed form beyond DS_6 .

0.3 2. The one stated condition: the dictionary

The theorem identifies the campaign's measured object with a Degtyarev-Shimada module, and this identification is the single place the campaign's own construction meets the published one. It is stated as a condition, not hidden.

The dictionary. The campaign's CRT all-A block at depth l is the Degtyarev-Shimada core module $\tilde{C}_{\{J(2s)\}}(2s)$, $s = l-1$. Concretely: the A-idempotent e_A (projecting onto the $(t-1)$ -primary part over F_p) corresponds to the coordinate-zeroing $z_v = 0$ of the partial Fermat $X(2s) := W_s \cap \{z_{\{2s+2\}} = \dots = z_{\{n+1\}} = 0\}$ [2, §4.6]; the live cofactor e_B ($a_i \neq 1$) corresponds to the surviving coordinates of Γ_K [2, Def 1.3, Thm 1.4].

This was established earlier in the campaign and confirmed verbatim against the Degtyarev-Shimada text (Definition 1.3's " $a_i \neq 1$ " is e_B ; " $a_i = 1$ " is e_A ; the partial-Fermat cut $X(2s)$ of §4.6 is the coordinate-zeroing). The theorem holds **given** this dictionary. A referee will examine exactly this link; the paper names it rather than assuming it.

0.4 3. Link 1 — the value is a closed walk (Eslabón 4)

This is the content of [THE_CLOSED_WALK_LAW.md](#), summarized for self-containment.

Proposition 1. $DS_n(m)$ is the number of closed walks of $n+2$ steps on $Z^{\lfloor (m-1)/2 \rfloor}$ (each step ± 1 along an axis, the walk returning to its origin), with a rest step permitted iff m is even. Equivalently it is the constant term of $(1 + S)^{n+2}$ (even m) or S^{n+2} (odd m), $S = x_{-1} + \dots + x_h + x_h^{-1} + \dots + x_{-1}^{-1}$ the symmetric Laurent sum.

This is read directly from [2, Remark 4.4]. It is gate-verified byte-exact on 24 in-sample points (DS_2, DS_4, DS_6 for $m = 3..8$), 12 out-of-sample points ($m = 9..12$), the new row DS_8 , and — the decisive check — the value $DS_6(4) = 1107$ measured independently on the author's hardware and reproduced by the formula without having been fitted to it. The parity correction δ_m is exactly the walker's rest permission: odd m no rest, even m rest allowed (Eslabón 4, [THE_CLOSED_WALK_LAW.md](#), §4).

With $s = l-1$, $m = d+1$, the depth- l value $DS_{\{2(l-1)\}}(d+1)$ is a closed walk of $2l$ steps on $Z^{\lfloor d/2 \rfloor}$, rest iff $m = d+1$ even, i.e. **iff d is odd**.

0.5 4. Link 2 — the measured block is the core, and the core is the walk (Eslabón 5, the induction)

This is the new content. The equality is carried by a tensor identity, **not** by a shared recursion (which would not force equality — two sequences being holonomic does not make them equal).

Lemma 2 (the block rank is the core dimension). Under the dictionary of §2, the rank of the measured block at depth l equals $\dim_C \tilde{C}_{\{J(2s)\}}(2s)$, $s = l-1$.

Proof. By [2, Cor 1.7], the relevant module decomposes as a tensor

$$\tilde{C}_{\{J_s\}}(2d) = \tilde{C}_{\{J(2s)\}}(2s) \otimes_Z \tilde{S}(s, d), \quad \tilde{S}(s, d) = \otimes_k Z[t_k]/(\phi(t_k)),$$

each tail factor $Z[t_k]/(\phi(t_k))$ free of rank $m - 1$, hence $\tilde{S}$ free. Degtyarev–Shimada obtain this by fixing the partition tail (the “constant relations” $t_{2s+2}t_{2s+3} = \dots = 1$, [2, §4.6]) and taking the even-index variables out. The exact sequence $0 \rightarrow (\lambda)/(\lambda\rho_J) \rightarrow R/(\lambda\rho_J) \rightarrow R/(\lambda) \rightarrow 0$ with $R/(\lambda)$ free [2, Lemma 4.5, Lemma 4.1] makes the split clean. Because $\tilde{S}$ is free of rank $(m-1)^{d-s}$, over $\mathbb{C}$ the tensor dimension factorizes as $\dim_{\mathbb{C}} \tilde{C}_{\{J_s\}}(2d) = \dim_{\mathbb{C}}(\text{core}) \cdot (m-1)^{d-s}$, and the tail contributes only its (constant, free) multiplicity. The rank the block contributes at depth l is therefore $\dim_{\mathbb{C}}$ of the core exactly — a free-module dimension, isolated, not a quantity that merely “advances controllably”. $\square$

Lemma 3 (the core dimension is the walk). $\dim_{\mathbb{C}} \tilde{C}_{\{J(2s)\}}(2s)$ is the closed walk of $2s+2$ steps of Proposition 1.

Proof. The core $\tilde{C}_{\{J(2s)\}}(2s)$ is the full-partition module one dimension down: by [2, §4.6], the restriction of J_s to the index set $\{0, \dots, 2s+1\}$ is exactly the full set $J(2s)$ of partitions of $2s+1$. Hence Remark 4.4 applies to the core verbatim with $n \rightarrow 2s$, and $\dim_{\mathbb{C}}(\text{core}) = DS_{\{2s\}}(m) = \text{the constant term of } (1 + S)^{2s+2} = \text{the closed walk}$. $\square$

Lemma 4 (the step is uniform in d). The depth step $l-1 \rightarrow l$ adds two lattice coordinates (Cor 1.7’s tail growth, the “+2 steps”) and is a finite-order holonomic operator on the walk count for every lattice dimension h .

Proof. By [4, Lipshitz 1988], the constant term of the N -th power of a fixed Laurent polynomial is P -recursive (holonomic) in N . The walk counts $cw(h, N)$ and $cwr(h, N)$ are exactly such constant terms with S the fixed symmetric Laurent polynomial of Z^h ; hence for each fixed h the sequence in N satisfies a finite-order linear recursion with polynomial coefficients. The depth step advances $N \rightarrow N+2$. The order is 1 for $h = 1$ (no rest), 2 for $h = 1$ (rest) and $h = 2$, and grows with h , but Lipshitz guarantees finiteness for every h — the uniformity of the step. This carries the *uniformity in d* ; it does **not** carry the equality, which is Lemmas 2–3. $\square$

Proof of the Theorem. Fix d, l . By Lemma 2, $\text{rec}(d, l) = \dim_{\mathbb{C}} \tilde{C}_{\{J(2s)\}}(2s)$, $s = l-1$. By Lemma 3, $\dim_{\mathbb{C}} \tilde{C}_{\{J(2s)\}}(2s) = \text{the closed walk of } 2l \text{ steps on } Z^{\lfloor d/2 \rfloor}$. By Proposition 1 (link 1), that walk is $DS_{\{2(l-1)\}}(d+1)$. Hence $\text{rec}(d, l) = DS_{\{2(l-1)\}}(d+1)$, standalone at each depth. The base $\text{rec}(d, 1) = d$ [Eslabón 1: the single-B-leg block rank is $\dim_B$, verified byte-exact for $d = 3, 4, 8$] anchors the ladder; Lemma 4 makes the depth step a finite uniform holonomic operator, so the identity propagates to all depths. $\square$

The equality rests on the tensor (Cor 1.7) and the generating count (Remark 4.4) — module-dimension identities from the source — with Lipshitz supplying only the finiteness of the step. No appeal to “two sequences share a recursion” is made or needed.

0.6 5. Verification

Every numeric instance is recomputed from scratch (see [chuchipachielconquistador.py](#) for link 1 and the block-rank ladders below).

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